

MICHAL SHLAPENTOKH-ROTHMAN

michal5@illinois.edu — michalmsr.web.illinois.edu — Google Scholar

EDUCATION

University of Illinois at Urbana-Champaign

Urbana, IL

PhD Candidate in Computer Science (AI Concentration)

Aug. 2020 – July 2026

Advisors: Derek Hoiem, Yuxiong Wang

Massachusetts Institute of Technology

Cambridge, MA

MEng & BS in Computer Science and Engineering

BS: 2015–2019; MEng: 2019–2020

MEng Thesis: Unifying Threat Data with Public Knowledge

RESEARCH EXPERIENCE

University of Illinois at Urbana-Champaign

Urbana, IL

Graduate Researcher

Fall 2020 – May 2026

- **Tool-based keyframe retrieval for long videos:** Developed a planner LLM that decomposes queries into tool calls and merges per-tool rankings with boolean operators, avoiding per-frame captioning. Improved the planner with GRPO and created a new benchmark of time-anchored questions for direct retrieval evaluation.
- **Data-efficient distillation for visual agents:** Designed template-based synthetic augmentation to distill visual tool-use behaviors into small coding LMs ($\leq 1\text{B}$ params), reducing reliance on expensive program annotations while improving multiple evaluation signals [1]
- **Region-Based Representations:** Integrated class-agnostic segmentation (SAM) with self-supervised features (e.g., DINOv2) to create compact region representations for downstream tasks such as semantic segmentation and object retrieval. [2]
- **Search + reflection for agents:** Integrated tree-search and self-reflection into LLM-based system for reasoning and planning tasks. [4]
- **Web interaction agents:** Developed an early LLM prompting approach that represents webpages as DOM-structured inputs and uses iterative exploration with environment feedback to complete web tasks. [3]

Amazon

Virtual

Applied Science Intern, Manager: Greg Hager, Mentor: Mohsen Malmir

May 2022 – Aug. 2022

- Investigated unsupervised category discovery in image classification tasks.
- Conducted initial implementations and proposed next steps.

Amazon

Virtual

Applied Science Intern, Manager: Greg Hager, Mentor: Ejaz Ahmed

May 2021 – Aug. 2021

- Investigated image classification transfer-learning under limited supervision.
- Conducted initial implementations and proposed next steps.

PUBLICATIONS

-
- [1] **M. Shlapentokh-Rothman**, P. Garg, Y.-X. Wang, and D. Hoiem, “Decomposing queries into tool calls for long-video keyframe retrieval,” in *In Submission*, 2026.
 - [2] **M. Shlapentokh-Rothman**, Y.-X. Wang, and D. Hoiem, “Visual program distillation with template-based augmentation,” *EMNLP (Findings)*, 2025.
 - [3] **M. Shlapentokh-Rothman***, A. Blume*, Y. Xiao, Y. Wu, S. TV, H. Tao, J. Y. Lee, W. Torres, Y.-X. Wang, and D. Hoiem, “Region-based representations revisited,” in *CVPR*, 2024.
 - [4] H. Tao, S. T V, **M. Shlapentokh-Rothman**, T. Gupta, H. Ji, and D. Hoiem, “WebWISE: Unlocking web interface control for LLMs via sequential exploration,” in *NAACL (Findings)*, Jun. 2024.
 - [5] A. Zhou, K. Yan, **M. Shlapentokh-Rothman**, H. Wang, and Y.-X. Wang, “Language agent tree search unifies reasoning acting and planning in language models,” in *ICML*, 2024.

- [6] D. Hoiem, T. Gupta, Z. Li, and **M. Shlapentokh-Rothman**, “Learning curves for analysis of deep networks,” in *ICML*, 2021.

SKILLS

PyTorch, Python, NumPy, Git

SERVICE

Reviewer CVPR, NeurIPS, ICLR, ICML, ARR, ECCV	2022 – Present
UIUC Vision Cluster Student Administrator	2022 – Present
UIUC Vision Mini-Conference Co-Organizer	April 2023

TEACHING EXPERIENCE

Applied Machine Learning UIUC CS 441, Graduate TA	Fall 2025
Computational Photography UIUC CS 445, Graduate TA	Spring 2021, 2023
Artificial Intelligence UIUC CS 440, Graduate TA	Fall 2020